

TRINIDAD STOKES, CO, USA

WMO: 724645

Lat: 37.262N

Lon: 104.338W

Elev: 5741

StdP: 11.89

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23070

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	3.7	9.2	-7.1	4.7	25.7	-2.3	6.0	30.1	31.9	46.8	26.8	46.9	6.8	270	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.0	94.0	59.3	91.5	59.1	89.4	59.1	64.4	81.0	63.2	80.5	62.2	79.7	10.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	96.6	67.7	58.4	90.3	67.4	56.8	85.3	67.3	32.5	80.2	31.5	80.0	30.7	79.9	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.7	24.9	20.8	DB	-7.3	98.4	6.7	2.1	-12.1	99.9	-16.0	101.1	-19.8	102.3	-24.6	103.8	(4)
(5)			WB	-7.9	66.7	6.5	1.6	-12.6	67.9	-16.4	68.8	-20.1	69.7	-24.8	70.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.8	33.3	36.0	43.5	49.9	59.4	70.0	74.5	72.4	65.7	53.3	42.1	33.0
	DBStd	17.19	10.36	10.80	10.11	9.45	8.17	6.59	4.70	4.43	7.53	9.37	10.06	10.47
	HDD50	2178	520	400	245	116	19	0	0	0	6	73	268	531
	HDD65	5285	983	812	667	453	206	28	2	4	80	370	687	993
	CDD50	3218	3	8	43	114	310	601	758	694	477	175	32	3
	CDD65	850	0	0	0	2	33	179	295	233	101	7	0	0
	CDH74	10660	0	1	20	106	720	2443	3318	2497	1311	235	9	0
	CDH80	4567	0	0	0	10	209	1132	1626	1085	469	36	0	0
Wind	WSAvg	9.9	9.5	9.9	10.7	11.6	10.8	10.6	9.4	9.0	9.4	9.5	9.4	9.1
Precipitation	PrecAvg	13.5	0.4	0.5	0.9	1.1	1.5	1.5	2.2	2.0	1.2	0.9	0.7	0.6
	PrecMax	20.7	1.3	1.3	2.6	4.6	4.9	6.4	6.5	5.0	3.0	2.8	2.4	1.8
	PrecMin	8.7	0.1	0.0	0.0	0.0	0.1	0.2	0.4	0.2	0.0	0.0	0.0	0.0
	PrecStd	3.4	0.3	0.4	0.6	1.1	1.1	1.3	1.5	1.3	0.7	0.8	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.8	70.4	76.8	81.3	89.7	97.2	97.0	94.8	91.5	84.0	75.2	66.1
		MCWB	43.4	44.7	47.5	49.9	55.1	58.0	60.8	59.6	58.0	53.1	47.8	43.7
	2%	DB	61.1	64.7	72.4	77.1	85.5	93.5	94.3	92.0	88.5	80.3	70.5	60.8
		MCWB	41.5	42.5	45.8	48.4	53.9	57.5	60.3	59.7	57.3	51.7	46.2	41.1
	5%	DB	56.0	60.7	68.1	73.5	81.7	90.5	91.9	89.5	85.6	76.1	65.8	55.9
		MCWB	39.0	40.9	44.4	47.6	53.1	57.2	60.3	60.1	56.8	51.0	44.3	39.3
	10%	DB	51.0	55.0	63.4	69.9	78.1	87.2	89.5	86.6	82.2	71.8	60.8	50.8
		MCWB	36.7	38.2	42.7	46.5	51.9	57.1	60.4	60.1	56.6	49.7	42.3	36.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.0	45.8	49.1	52.9	59.6	63.7	66.1	65.9	62.6	58.0	49.3	44.8
		MCDB	64.1	67.6	71.8	73.0	78.6	83.4	81.7	80.5	78.5	74.4	70.6	64.0
	2%	WB	41.8	43.3	46.9	50.5	57.3	61.9	64.6	64.3	60.9	55.1	47.2	42.0
		MCDB	59.3	63.6	69.0	70.7	76.3	81.4	81.4	80.2	77.8	72.0	67.9	58.9
	5%	WB	39.5	41.2	45.2	49.0	55.5	60.5	63.5	63.1	59.6	52.6	45.2	39.7
		MCDB	54.8	59.0	66.0	68.8	73.7	80.2	81.1	79.6	77.3	70.7	63.8	55.1
	10%	WB	37.1	38.9	43.3	47.4	53.9	59.3	62.5	62.0	58.4	50.6	42.9	37.1
		MCDB	50.3	54.1	61.8	67.5	72.7	79.3	80.4	78.7	76.0	68.8	59.7	50.5
Mean Daily Temperature Range	5% DB	MDBR	26.7	27.7	29.2	28.2	28.4	29.4	28.0	27.1	27.8	28.9	28.4	26.4
		MCDBR	33.4	34.8	35.2	34.2	34.1	33.3	31.2	30.7	31.3	34.0	35.0	33.8
		MCWBR	18.3	18.0	16.4	14.8	13.6	11.8	8.9	9.2	10.5	14.1	16.8	18.6
	5% WB	MCDBR	31.8	33.4	33.5	31.3	29.7	28.9	26.8	26.4	26.6	29.7	32.7	32.1
		MCWBR	18.1	17.5	16.3	14.4	12.7	11.3	8.9	9.3	9.4	13.1	16.1	18.1
Clear-Sky Solar Irradiance	taub		0.236	0.242	0.269	0.293	0.307	0.327	0.359	0.343	0.303	0.273	0.248	0.239
	taud		2.629	2.632	2.547	2.481	2.503	2.451	2.435	2.466	2.546	2.608	2.605	2.606
	Ebn at Noon		313	322	319	314	309	300	290	293	302	305	305	304
	Edh at Noon		22	25	30	34	34	36	36	34	30	25	22	21
All-Sky Solar Radiation	RadAvg		945	1236	1638	1984	2234	2379	2242	2036	1784	1385	1032	824
	RadStd		62	87	110	118	142	97	91	86	93	78	67	49

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.79	N/A	-2.77	+1.26	N/A	N/A	N/A	N/A	+187	+135
(47) Regional (16 neighbors)		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+77

Nomenclature: See separate page